

# FPTR: Feasibility-Preserving Transactional Refinement for Deadline-Constrained Joint Beam and Resource Allocation: Appendix

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## A Evidence and Reproducibility Summary

This appendix summarizes execution, stability, stage acceptance, exact calibration, and artifact integrity; quality uses five-run instance medians, and counts denote separate launches.

The appendix is a companion record rather than a second results narrative. It makes the evaluation unit, crediting rule, and audit boundary explicit. Each reported quantity can therefore be traced to a process launch, a generated case, or a sealed artifact.

### A.1 Reading Conventions and Statistical Units

A launch denotes one independent subprocess invocation with its own input stream, execution order, random seed, and wall-clock measurement. The validator reparses the serialized allocation and recomputes the transmitted-traffic objective. A trace-reported score is diagnostic and never replaces this independent recomputation.

For instance  $i$  and method  $m$ , let  $F_{i,m,j}$  be the credited traffic from repeat  $j$ . The quality unit used in the tables is

$$\tilde{F}_{i,m} = \text{median}_{j=1,\dots,5} F_{i,m,j}, \quad g_i(A, B) = \frac{\tilde{F}_{i,A} - \tilde{F}_{i,B}}{\tilde{F}_{i,B}}. \quad (1)$$

Relative gains are averaged within each scenario before the five scenarios receive equal weight. This prevents the largest instances from determining the summary by case size alone. A late but legal output remains available for diagnosis, whereas its credited traffic is zero under the stated service-level agreement.

### A.2 Execution Accounting and Configuration-Level Results

**Table A1.** Formal internal launch accounting; valid outputs passed independent checks.

| Suite             | Instances | Conditions | Repeats | Launches | Valid | Misses |
|-------------------|-----------|------------|---------|----------|-------|--------|
| Main              | 150       | 6          | 5       | 4,500    | 4,500 | 0      |
| Budget sweep      | 150       | 4          | 5       | 3,000    | 3,000 | 0      |
| CG stress         | 50        | 3          | 5       | 750      | 750   | 0      |
| Exact calibration | 12        | 6          | 5       | 360      | 360   | 0      |
| Total             | —         | —          | —       | 8,610    | 8,610 | 0      |

The internal main and budget suites reused 150 inputs but launched each condition independently; the 500-ms harness assigned zero utility to nonzero exits, illegal outputs, or wall time above 100 ms.

The four internal suites serve different purposes. The main suite measures cumulative configuration behavior. The budget sweep changes the search cap while retaining the same evaluation family. The CG stress suite increases group size and coverage together. The exact suite adds independent optimization on tiny cases alongside the heuristic runs. Their launch counts should therefore not be interpreted as four disjoint collections of inputs.

### A.3 Matched External-Baseline Suite

The extended external main suite used the same 150 held-out inputs, five repeats, validator, objective, 87-ms budget, and 100-ms SLA for Full and six independent baselines: ALNS, tabu search, GA, SA, ILS, and GRASP. It contained 6,750 main-suite launches. Every output was valid; 219 external launches exceeded the SLA. The earlier budget and exact calibration remain archived under `../reproducibility/external_comparison_extended/`.

The external programs share the input and output contract, objective, and validation procedure, but they do not call FPTR’s core scheduler or reuse its stage traces. Each method is compiled as an independent C++17 executable. Thus, the comparison tests independently written reference implementations under a common accounting rule, rather than claiming to reproduce every published solver in the literature.

**Table A2.** Full versus six independent external baselines at the matched 87-ms main condition.

| Baseline | W/T/L    | Mean rel. |            | Main misses/<br>750 runs |
|----------|----------|-----------|------------|--------------------------|
|          |          | gain (%)  | 95% CI (%) |                          |
| ALNS     | 134/5/11 | 7.61      | 7.03–8.56  | 74                       |
| Tabu     | 139/4/7  | 7.75      | 7.01–8.52  | 18                       |
| GA       | 127/2/21 | 6.58      | 5.91–7.47  | 44                       |
| SA       | 137/4/9  | 7.90      | 7.19–8.58  | 15                       |
| ILS      | 137/4/9  | 8.40      | 7.53–9.13  | 41                       |
| GRASP    | 140/2/8  | 7.59      | 6.88–8.33  | 27                       |

Relative gains use nonzero baseline medians: 141/150 ALNS pairs, 149/150 tabu pairs, 144/150 GA pairs, 150/150 SA pairs, 148/150 ILS pairs, and 149/150 GRASP pairs. W/T/L counts include all 150 instance medians; the machine-readable archive is `../reproducibility/external_comparison_v3/`.

**Table A3.** Main-set quality and wall time over 150 five-run instance medians.

| Configuration | Mean $\tilde{F}$ | Rel. to Base (%) | Wall p50 (ms) | Wall p95 (ms) | Worst (ms) | Misses/<br>runs |
|---------------|------------------|------------------|---------------|---------------|------------|-----------------|
| BeamFirst     | 13467.89         | -1.82            | 4.23          | 7.71          | 8.82       | 0/750           |
| Base          | 13662.15         | +0.00            | 15.75         | 22.68         | 31.50      | 0/750           |
| Global        | 13732.17         | +0.44            | 27.06         | 62.12         | 64.40      | 0/750           |
| CG            | 13796.05         | +0.88            | 46.40         | 73.30         | 78.09      | 0/750           |
| Remask        | 13873.17         | +1.54            | 51.44         | 81.35         | 82.85      | 0/750           |
| Full          | 13898.13         | +1.79            | 60.28         | 87.29         | 92.14      | 0/750           |

Full led mean instance-median traffic, with 60.28/87.29/92.14-ms p50/p95/worst time and no miss in 750 launches. Table A3 weights regimes equally and pools 750 wall times; it describes quality–runtime rather than compute-matched superiority.

Wall time includes process launch, input transfer, solver execution, and output collection. Independent Python validation is reported separately. The 87-ms value is a search budget, not a hard completion-time guarantee, so the percentiles describe the observed host and implementation rather than a platform-independent bound.

### A.4 Run-Level Stability and Stage Acceptance

Across 750 same-repeat pairs, Full versus Base yielded 564/186/0 wins/ties/losses (1.800% mean gain), and Full versus BeamFirst yielded 643/107/0 (3.839%). Scores were repeat-invariant for 122/150 instances; the relative five-run range had 4.969% p95 and 7.592% maximum.

The repeated-process comparison and the stage-acceptance trace answer different questions. Repeated processes quantify outcome stability under ordinary scheduling noise. The trace records the fixed order Base–Global–CG–Remask–Pair inside one Full run, so each acceptance count is conditional on the incumbent produced by earlier stages.

Overall instance/run acceptance was 48.0/48.0% (Global), 22.7/23.3% (CG), 26.0/26.0% (Remask), and 26.7/26.7% (Pair). A 0/0 denotes no strict improvement because of a tie, expiry, or validation failure; counts are fixed-order conditional increments.

**Table A4.** Fixed-order stage acceptance by scenario (instances/runs).

| Scenario          | $n/5n$  | Global | CG     | Remask | Pair   |
|-------------------|---------|--------|--------|--------|--------|
| small-balanced    | 30/150  | 2/10   | 8/40   | 5/25   | 3/15   |
| medium-longtail   | 30/150  | 18/90  | 11/55  | 17/85  | 23/115 |
| medium-tight      | 30/150  | 7/35   | 3/15   | 17/85  | 14/70  |
| large-mixed       | 30/150  | 27/135 | 10/49  | 0/0    | 0/0    |
| large-nonadjacent | 30/150  | 18/90  | 2/16   | 0/0    | 0/0    |
| Overall           | 150/750 | 72/360 | 34/175 | 39/195 | 40/200 |

These percentages are not standalone module success rates. A later stage may have few accepted candidates because earlier stages already removed the corresponding residual structure, because the remaining budget is short, or because the candidate fails the common gate. The cumulative Full score can still improve when accepted changes are sparse, provided that each accepted change is feasible and strictly better.

### A.5 Exact Calibration and Independent Audit

**Table A5.** Exact calibration; BF=BeamFirst; W01/C02–C12=seeds 20281001–20281012.

| Case | $F^*$ | BF  | Base | Full | Gap   | Case | $F^*$ | BF   | Base | Full | Gap  |
|------|-------|-----|------|------|-------|------|-------|------|------|------|------|
| W01  | 920   | 690 | 690  | 690  | 25.00 | C07  | 449   | 449  | 449  | 449  | 0.00 |
| C02  | 237   | 237 | 237  | 237  | 0.00  | C08  | 806   | 672  | 776  | 776  | 3.72 |
| C03  | 548   | 548 | 548  | 548  | 0.00  | C09  | 804   | 528  | 744  | 744  | 7.46 |
| C04  | 1188  | 908 | 1188 | 1188 | 0.00  | C10  | 1665  | 1589 | 1631 | 1631 | 2.04 |
| C05  | 564   | 564 | 564  | 564  | 0.00  | C11  | 888   | 888  | 888  | 888  | 0.00 |
| C06  | 648   | 648 | 648  | 648  | 0.00  | C12  | 888   | 888  | 888  | 888  | 0.00 |

Independent DP reproduced all 12 sealed optima without the formal solver: 390,814 attempted transitions became 332,585 pre-prune states after 58,229 duplicate collapses; dominance removed 219,088, leaving 113,497. Layer and aggregate identities held; this suite calibrates only small-case quality.

For each enumerated beam-mask plan, the audit dynamic program carries a vector of accumulated per-user traffic after every resource layer. It enumerates the empty choice and legal compatibility-group choices, clips each vector by user buffers, and removes component-wise dominated states. This construction is independent of the heuristic implementation and explains why the audited optimum is a useful calibration target rather than a second heuristic baseline.

### A.6 Budget, CG Stress, and Artifact Integrity

At 20/40/60/87 ms, gains over 20 ms were 0.00/1.03/1.60/1.65%, with p95 times 20.44/40.34/60.39/87.31 ms; all 750 launches per budget met 100 ms, and the 87-ms suite was separate. For CG maxima 2/5/10/15/20, coverage and Full–Base gain co-varied from 6.7%/0.11% to 66.7%/1.81% (intermediate gains: 0.75/3.23/2.56%); size 20 gave 87.50/92.93-ms p95/worst time and 0/50 misses. SHA-256 manifests bind 14 sealed-result, 16 exact-audit (12 reconstructed-input), and 21 evidence files. Hash-reconstructed inputs make only the optimizer’s path independent; **causal cumulative ablation** is a checksum-preserved fixed-order trace.

The integrity chain is layered. Case hashes bind generated inputs, source hashes bind the generator and validator versions, and directory checksums bind the derived result files. A regenerated archive is therefore auditable at both the numerical and provenance levels. The appendix does not treat retained binaries as evidence; the documented build commands and compiler flags are the reproducible boundary.